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F78AB
Actuarial and Financial Mathematics

Chapter 7: Loans and Loan Schedules

LEADERS IN IDEAS AND SOLUTIONS

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Three Ways to Repay a Loan

Say you borrow £L from a financial institution and have to repay the loan in n years. This loan can be repaid in 3 ways:

1. Repay the loan and all the interest [$L(1+i)^n$] at the end of the term.
2. Pay interest only (iL) during the term and repay the principal (L) at the end of the term
3. Repay level amounts (£X), where the amount, £X, is calculated so that the entire loan is repaid with the final instalment of £X.

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An Example

Suppose you borrow £10,000 over 4 years at an effective interest rate of 5% per annum and agree to make payments of £2,820.12 per annum at the end of each year.

Time (Yrs)	Repayment	Interest Content	Capital Content	Capital Outstanding
0				10,000.00
1	2820.12	500.00	2320.12	7679.88
2	2820.12	383.99	2436.13	5243.75
3	2820.12	262.19	2557.93	2685.82
4	2820.12	134.29	2685.83	-0.01

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The Principle of Equivalence

Consider the following 2 sets of cashflows:

1. $c_1, c_2, c_3, \ldots c_n$ payable at times $t_1, t_2, t_3, \ldots, t_n$
2. $d_1, d_2, d_3, \ldots d_m$ payable at times $s_1, s_2, s_3, \ldots, s_m$

We say that the two sets of cashflows are equivalent at time t if we can find some t at which they have the same value i.e.

$$\sum_{r=1}^{n} c_r V(t, t_r) = \sum_{r=1}^{m} d_r V(t, s_r)$$

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The Principle of Equivalence

- If 2 sets of cashflows are equivalent at any time t then they are also equivalent at any other time s,

$$\sum_{r=1}^{n} c_r V(s, t_r) = \sum_{r=1}^{m} d_r V(s, s_r)$$

- We can drop the specification of a time t from the definition, and just say that the two sets of cash flows are equivalent.
- Note that the two sets of cashflows will generally **not be equivalent** under a different interest rate regime.

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The Principle of Equivalence

- Often, one of the sets of cash flows is a single cashflow payable now, and the other is a series of one or more cashflows payable in the future.
- We regard the single cashflow payable now as the price of the cashflows payable in the future.

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The Principle of Equivalence

- **Principle of Equivalence**

We assume that any person or entity is indifferent about which of two equivalent series of cashflows they own. That is, they would be willing to exchange one set of cashflows for any other equivalent set of cashflows.

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The Principle of Equivalence

- **Some comments**
  - It is logical to suppose that the person or entity would prefer to own a set of cashflows with a higher value.
  - We can use the **Principle of Equivalence** to find the fair price of a financial asset, that is the lump sum that you pay at the outset in exchange for a future stream of cashflows.

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The Principle of Equivalence

Example

A bank lends you £10,000 for 4 years. The effective rate of interest is 5% per annum. Show that the following three ways of repaying the loan are equivalent:

1. A single repayment of £12,155.06 at the end of the term.
2. A payment of £500 at the end of each of the next 4 years, and repayment of the £10,000 at the end of 4 years.
3. Level annual repayments of £2,820.12 paid yearly in arrears.

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The Principle of Equivalence

You have been offered an investment that will pay you £200 at the end of each of the next 24 months. What is the maximum price you should pay if the effective rate of interest is 0.5% per month?

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The Mathematics of Loan Schedules

By the Principle of Equivalence, the present value of a loan of amount *L* is equivalent to a future series of repayments, made in arrears, of *X* per time unit for a term of n time units at an effective rate of interest of i per time unit and is given by

$$L_0 = L = X a_{\overline{n}|}^{i}$$

The Mathematics of Loan Schedules

Read Notes (chapter 7)

- By working through the loan schedule, the following results can be shown:
  - The capital outstanding just after the rth repayment is

$$L_r = X a_{\overline{n-r}|}^{i}$$

  - The interest content of the rth repayment is

$$i \times L_{r-1} = iX a_{\overline{n+1-r}|}^{i}$$

  - The capital content of the rth repayment is

$$X - iX a_{\overline{n+1-r}|}^{i} = X\left(1 - i\left(\frac{1 - v^{n+1-r}}{i}\right)\right) = Xv^{n+1-r}$$

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The Mathematics of Loan Schedules

Example

For the 4 year loan in above (£10,000 repayable over 4 years at 5% per annum effective), using the results in the previous slide verify the following.

1. The capital outstanding after the 1st repayment is £7,679.88.
2. The capital content of the 2nd repayment is £2,436.13.
3. The interest content of the 3rd repayment is £262.19.

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The Mathematics of Loan Schedules

Example (You try)

A 25-year loan of amount £200,000 is issued on the basis of an effective rate of interest of 4% per annum. Repayments are to be made annually in arrears.

1. What's the capital outstanding immediately after the 11th repayment is made?
2. What is the capital content of the 11th repayment?
3. What is the interest content of the 11th repayment?

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Changing the Terms of a Loan

It is common for a borrower to ask for a change in the term of a loan, for example:-

- to extend or shorten the loan period,
- to miss one or more payments,
- to repay part of the loan early.

The lender will then need to calculate a new repayment amount by

- calculating the outstanding capital at the time of the requested change,
- then calculate the new repayment amount to repay the calculated amount of outstanding capital.

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Changing the Terms of a Loan

Example

A borrower takes out a loan of £70,000 to be repayable by level annual repayments over 20 years, where the repayments are calculated at an effective rate of interest of 7% per annum.

1. How much is each annual instalment?
2. What is the new repayment amount if the borrower asks for the term to be extended by five years, immediately after making the 13th repayment?

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Changing the Terms of a Loan cont..

Example cont…

A borrower takes out a loan of £70,000 to be repayable by level annual repayments over 20 years, where the repayments are calculated at an effective rate of interest of 7% per annum.

3. If instead of extending the term the borrower had asked to miss the 14th and 15th repayments, what would each of the remaining 5 repayments be?
4. What would the new repayment amount become if the borrower decided to repay an additional £5,000 at time 13 (on top of the standard repayment), with no change to the term of the loan?

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Changing the Interest Rate

Often more than one interest rate is involved during the term of the loan. This may be because:

– Interest rates were planned to change during the term, for example, the lender charged a lower-than-normal rate of interest for the first few years.

– The lender changes the interest rate for all borrowers to reflect changed market conditions.

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Changing the Interest Rate

Example

You borrow £100,000 over 30 years to be repaid by level annual repayments. The rate of interest will be 6% per annum effective for the first 15 years and 7% per annum effective thereafter. Find the repayment amount.

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Changing the Interest Rate

Another example

You borrow £50,000 from a bank, to be repaid over 25 years by level annual repayments based on an effective rate of interest of 3% per annum. Just after the 5th repayment has been paid, the bank raises its interest rate to 5% per annum effective. What is the new repayment?

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Non-Level Repayments

- There is no need for repayments to be level.
- No new principles are involved.
- Will need to work things out from first principles.
- See Tutorial 5 Questions 8 and 9.